

ANALYYSI I: LIMITS

JOONA PIIRAINEN

ABSTRACT. Notes for the University of Helsinki course MAT11003, taught in Finnish from 01.08.2026–17.10.2026.

CONTENTS

1.	Course information	1
2.	Lecture notes	1
2.1.	01.09.2026	1
2.2.	03.09.2026	2
2.3.	08.09.2026	4
2.4.	10.09.2026	6
2.5.	15.09.2026	10
2.6.	17.09.2026	11
2.7.	22.09.2026	14

1. COURSE INFORMATION

The official course information is available on the University of Helsinki course page.

2. LECTURE NOTES

2.1. 01.09.2026.

2.1.1. *The absolute value function.*

Definition 2.1. For any real number $x \in \mathbb{R}$, the *absolute value* (or *modulus*) of x , denoted by $|x|$, is defined by

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

Geometrically, $|x - y|$ is the distance between the points x and y on the real line.

Example 2.2. Fix $y \in \mathbb{R}$ and let $h > 0$. The inequality $|x - y| < h$ can be written in several equivalent ways:

$$\begin{aligned} |x - y| < h &\iff -h < x - y < h \\ &\iff y - h < x < y + h \\ &\iff x \in (y - h, y + h). \end{aligned}$$

Thus, $|x - y| < h$ means that x lies in the open interval of radius h centred at y .

Example 2.3. As a concrete instance of the preceding equivalences,

$$\begin{aligned} |2x - 1| < 3 &\iff -3 < 2x - 1 < 3 \\ &\iff -2 < 2x < 4 \\ &\iff -1 < x < 2 \\ &\iff x \in (-1, 2). \end{aligned}$$

Theorem 2.4 (Triangle and reverse triangle inequalities). *For all $x, y \in \mathbb{R}$,*

$$\begin{aligned} |x + y| &\leq |x| + |y|, \\ ||x| - |y|| &\leq |x - y|. \end{aligned}$$

Proof. By definition 2.1,

$$\begin{aligned} -|x| &\leq x \leq |x|, \\ -|y| &\leq y \leq |y|. \end{aligned}$$

Adding the inequalities gives

$$-(|x| + |y|) \leq x + y \leq |x| + |y|,$$

which is equivalent to $|x + y| \leq |x| + |y|$.

Applying this triangle inequality to $x = (x - y) + y$ gives

$$|x| - |y| \leq |x - y|.$$

Interchanging x and y similarly gives $|y| - |x| \leq |x - y|$. Therefore

$$-|x - y| \leq |x| - |y| \leq |x - y|,$$

which is equivalent to the reverse triangle inequality. $\square$

2.2. 03.09.2026.

2.2.1. More on absolute values.

Theorem 2.5. *We have*

$$|xy| = |x||y|$$

for all $x, y \in \mathbb{R}$.

Proof. We prove by splitting into four cases. Firstly, if $x, y \geq 0$, we have $xy \geq 0$, so $|xy| = xy$. But in this case also both $|x| = x$ and $|y| = y$, so $|xy| = xy = |x||y|$. Secondly, if $x, y < 0$, we have $xy > 0$, $|xy| = xy = (-x)(-y) = |x||y|$. Now suppose $x < 0$ and $y \geq 0$. Then $xy \leq 0$, and hence $|xy| = -xy = (-x)y = |x||y|$. The case $x \geq 0$, $y < 0$ follows from commutativity of multiplication of real numbers. $\square$

Example 2.6. Let $x \in (1, 3)$ and $y \in (4, 8)$. Find an upper bound for

$$|x + y - 8|.$$

Kolmioepäyhtälön nojalla saadaan

$$\begin{aligned} |x + y - 8| &= |(x - 2) + (y - 6)| \\ &\leq |x - 2| + |y - 6| \\ &< 1 + 2 \\ &= 3. \end{aligned}$$

Thus we have found that 3 is an upper bound for $|x + y - 8|$.

Example 2.7. Let $x \in (2, 6)$ and $y \in (2, 8)$. Find an upper bound for

$$|xy - 20|.$$

We have $|x - 4| < 2$ and $|y - 5| < 3$. We have

$$\begin{aligned} |xy - 20| &= |xy - 4y + 4y - 20| \\ &\leq |xy - 4y| + |4y - 20| \\ &= |y||x - 4| + 4|y - 5| \\ &\leq 8 \cdot 2 + 4 \cdot 3 \\ &= 28. \end{aligned}$$

It follows that 28 is an upper bound for $|xy - 20|$.

Example 2.8. Find a positive real number m such that

$$|x^2 - 16| \leq m|x - 4|,$$

where $x \in (0, 10)$.

When $x = 4$, we get $0 \leq 0$, which holds with every $m \in \mathbb{R}$. Now suppose $x \neq 4$. We simplify the inequality and obtain

$$|x + 4| \leq m.$$

Since $x \in (0, 10)$, $|x + 4| \in (4, 14)$. We have thus obtained

$$|x + 4| \leq 14,$$

so we can choose $m = 14$.

2.2.2. Field and ordering properties of $\mathbb{R}$.

Definition 2.9. $(F, +, \cdot)$ is a field if and only if

- (1) The algebraic structure $(F, +)$ is an *abelian group*.
- (2) The algebraic structure $(F^*, \cdot)$ is an *abelian group*, where $F^* = F \setminus \{0_F\}$.
- (3) The operation $\cdot$ distributes over $+$.

Theorem 2.10. $\mathbb{R}$ is a field.

Proof. Consider the usual addition and multiplication on $\mathbb{R}$.

First, $(\mathbb{R}, +)$ is an abelian group. Addition is closed, associative, and commutative; 0 is an additive identity; and every $x \in \mathbb{R}$ has the additive inverse $-x$.

Next, $(\mathbb{R} \setminus \{0\}, \cdot)$ is an abelian group. Multiplication is associative and commutative, and 1 is its identity element. If $x \neq 0$, then $x^{-1} = 1/x$ is a nonzero real number satisfying

$$x \cdot x^{-1} = 1.$$

The product of two nonzero real numbers is nonzero, so multiplication is closed on $\mathbb{R} \setminus \{0\}$.

Finally, multiplication distributes over addition:

$$x(y + z) = xy + xz$$

for all $x, y, z \in \mathbb{R}$. Therefore $(\mathbb{R}, +, \cdot)$ is a field. □

Definition 2.11 (Ordered field). An *ordered field* is a field $(F, +, \cdot)$ equipped with a relation $<$ such that, for all $x, y, z \in F$,

- (1) exactly one of $x < y$, $x = y$, or $x > y$ holds;
- (2) if $x < y$ and $y < z$, then $x < z$;
- (3) if $x < y$, then $x + z < y + z$; and
- (4) if $0_F < x$ and $0_F < y$, then $0_F < xy$.

Here $x > y$ means $y < x$. The first two conditions state that $<$ is a strict total order, while the last two state that the order is compatible with the field operations.

Theorem 2.12. *The field $(\mathbb{R}, +, \cdot)$, equipped with the usual order $<$, is an ordered field.*

Proof. We have already shown that $(\mathbb{R}, +, \cdot)$ is a field. The usual strict order on $\mathbb{R}$ satisfies trichotomy and transitivity, so it satisfies the first two ordering axioms.

Suppose that $x < y$. Then $0 < y - x$, and for every $z \in \mathbb{R}$,

$$(y + z) - (x + z) = y - x > 0.$$

Hence $x + z < y + z$. Furthermore, if $0 < x$ and $0 < y$, then the usual sign rule for multiplication gives $0 < xy$. Thus the order is compatible with addition and multiplication, and $\mathbb{R}$ is an ordered field. $\square$

Theorem 2.13 (Uniqueness of additive identity). *The identity element 0_F of the abelian group $(F, +)$ is unique.*

Proof. Suppose that 0_F and $0'_F$ are both additive identity elements of $(F, +)$. Since 0_F is an additive identity,

$$0_F + 0'_F = 0'_F.$$

Since $0'_F$ is also an additive identity,

$$0_F + 0'_F = 0_F.$$

Therefore $0_F = 0'_F$. $\square$

Theorem 2.14 (Cancellation law). *Let $x, y, z \in \mathbb{R}$. If $x + y = x + z$, then $y = z$.*

Proof. Using the additive identity, additive inverse, associativity, and the assumption $x + y = x + z$, we obtain

$$\begin{aligned} y &= 0 + y \\ &= ((-x) + x) + y \\ &= (-x) + (x + y) \\ &= (-x) + (x + z) \\ &= ((-x) + x) + z \\ &= 0 + z \\ &= z. \end{aligned}$$

$\square$

2.3. 08.09.2026.

2.3.1. Sequences.

Definition 2.15. Let X and Y be sets. A function $f: X \rightarrow Y$ is a relation $f \subseteq X \times Y$ such that, for every $x \in X$, there exists exactly one $y \in Y$ with $(x, y) \in f$. This unique y is denoted by $f(x)$.

The set X is the *domain* of f , and Y is its *codomain*. The set

$$f(X) = \{f(x) : x \in X\}$$

is its *range* or *image*.

Definition 2.16. A sequence of real numbers is a function $x: \mathbb{N} \rightarrow \mathbb{R}$. For $n \in \mathbb{N}$, the value $x(n)$, usually denoted by x_n , is the n -th term of the sequence. We denote the sequence by $(x_n)_{n \in \mathbb{N}}$, or simply by (x_n) .

2.3.2. Limit of a sequence.

Definition 2.17. Let (x_n) be a sequence of real numbers. We say that (x_n) *converges* to $a \in \mathbb{R}$ if, for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that, for every $n \in \mathbb{N}$ with $n > N$,

$$|x_n - a| < \epsilon.$$

In this case, we write $x_n \rightarrow a$ as $n \rightarrow \infty$, or $\lim_{n \rightarrow \infty} x_n = a$.

Let us look at some examples.

Example 2.18. Let (x_n) be defined by

$$x_n = \frac{2n}{n+1}.$$

We will show that

$$\lim_{n \rightarrow \infty} x_n = 2.$$

Let $\epsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \frac{2}{\epsilon}$. Now for every $n > N$, we have

$$\begin{aligned} |x_n - 2| &= \left| \frac{2n}{n+1} - 2 \right| \\ &= \frac{2}{n+1} < \frac{2}{n} < \frac{2}{\frac{2}{\epsilon}} = \epsilon. \end{aligned}$$

Thus $x_n \rightarrow 2$ when $n \rightarrow \infty$.

Example 2.19. Let (x_n) be defined by

$$\begin{aligned} x_0 &= \cdots = x_7 = 42, \\ x_n &= 7 + \frac{1}{n}, \quad n \geq 8. \end{aligned}$$

We will show that

$$\lim_{n \rightarrow \infty} x_n = 7.$$

Let $\epsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \max \left\{ 8, \frac{1}{\epsilon} \right\}$. Now for every $n > N$, we have

$$|x_n - 7| = \frac{1}{n} < \epsilon.$$

Thus $x_n \rightarrow 7$ as $n \rightarrow \infty$.

Let us now tackle a simple divergence proof.

Example 2.20. Let us show that the sequence x_n defined by $x_n = n$ diverges. Assume that the sequence converged to some $a \in \mathbb{R}$. Taking $\epsilon = 1$, there exists $N \in \mathbb{N}$ such that, for every $n > N$,

$$|x_n - a| < \epsilon = 1.$$

Choose $n \in \mathbb{N}$ such that

$$n > \max \{N, |a| + 1\}.$$

By the triangle inequality,

$$n = |n| = |(n - a) + a| \leq |n - a| + |a|.$$

Subtracting $|a|$ from both sides and using $n > |a| + 1$, we obtain

$$|x_n - a| = |n - a| \geq n - |a| > 1,$$

contradicting the assumption $|x_n - a| < 1$. Therefore (x_n) diverges.

Example 2.21. Let (x_n) be defined by

$$\begin{aligned} x_0 &= \cdots = x_{42} = 1, \\ x_n &= \frac{n}{n-42}, \quad n \geq 43. \end{aligned}$$

We will show that

$$\lim_{n \rightarrow \infty} x_n = 1.$$

Let $\epsilon > 0$. Choose $N \in \mathbb{N}$ such that

$$N > \max \left\{ 84, \frac{84}{\epsilon} \right\}.$$

Now, for every $n > N$, we have $n > 84$, and hence $\frac{n}{2} - 42 > 0$. Therefore

$$\begin{aligned} |x_n - 1| &= \left| \frac{n}{n-42} - 1 \right| \\ &= \frac{42}{n-42} \\ &= \frac{42}{\frac{n}{2} + \frac{n}{2} - 42} \\ &< \frac{84}{n} < \epsilon. \end{aligned}$$

Thus $x_n \rightarrow 1$ as $n \rightarrow \infty$.

2.4. 10.09.2026.

2.4.1. Uniqueness of limits.

Theorem 2.22 (Uniqueness of limits). *A convergent sequence of real numbers has a unique limit. That is, if $x_n \rightarrow a$ and $x_n \rightarrow b$ as $n \rightarrow \infty$, then $a = b$.*

Proof. Suppose for contradiction that $a \neq b$, and set

$$\epsilon = \frac{|a - b|}{2} > 0.$$

Since $x_n \rightarrow a$ and $x_n \rightarrow b$, there exist $N_a, N_b \in \mathbb{N}$ such that

$$n > N_a \implies |x_n - a| < \epsilon$$

and

$$n > N_b \implies |x_n - b| < \epsilon.$$

Choose $n > \max \{N_a, N_b\}$. By the triangle inequality,

$$|a - b| \leq |a - x_n| + |x_n - b| < 2\epsilon = |a - b|,$$

which is impossible. Therefore $a = b$. □

Theorem 2.23 (Convergent sequences are bounded). *Every convergent sequence of real numbers is bounded.*

Proof. Suppose that $x_n \rightarrow a$. Taking $\epsilon = 1$, there exists $N \in \mathbb{N}$ such that

$$n > N \implies |x_n - a| < 1.$$

By the triangle inequality, for every $n > N$,

$$|x_n| \leq |x_n - a| + |a| < 1 + |a|.$$

Let

$$M = \max \{|x_0|, |x_1|, \dots, |x_N|, 1 + |a|\}.$$

Then $|x_n| \leq M$ for every $n \in \mathbb{N}$, so (x_n) is bounded. □

2.4.2. Algebra of limits.

Theorem 2.24 (Algebraic limit laws). *Suppose that $x_n \rightarrow a$ and $y_n \rightarrow b$, and let $c \in \mathbb{R}$. Then*

- (1) $x_n \pm y_n \rightarrow a \pm b$.
- (2) $cx_n \rightarrow ca$.
- (3) $x_n y_n \rightarrow ab$.
- (4) If $b \neq 0$, then $y_n \neq 0$ for all sufficiently large n , and

$$\frac{x_n}{y_n} \rightarrow \frac{a}{b}.$$

Proof. (1) Let $\epsilon > 0$. Since $x_n \rightarrow a$ and $y_n \rightarrow b$, there exist $N_x, N_y \in \mathbb{N}$ such that

$$n > N_x \implies |x_n - a| < \frac{\epsilon}{2}$$

and

$$n > N_y \implies |y_n - b| < \frac{\epsilon}{2}.$$

Thus, for $n > \max\{N_x, N_y\}$,

$$|(x_n + y_n) - (a + b)| \leq |x_n - a| + |y_n - b| < \epsilon.$$

Hence $x_n + y_n \rightarrow a + b$. Replacing y_n and b by $-y_n$ and $-b$ gives $x_n - y_n \rightarrow a - b$.

- (2) If $c = 0$, then $cx_n = ca = 0$ for every n . If $c \neq 0$, choose $N \in \mathbb{N}$ such that

$$n > N \implies |x_n - a| < \frac{\epsilon}{|c|}.$$

Then, for every $n > N$,

$$|cx_n - ca| = |c||x_n - a| < \epsilon.$$

Therefore $cx_n \rightarrow ca$.

- (3) Since (y_n) converges, it is bounded. Choose $M > 0$ such that $|y_n| \leq M$ for every n . There exist $N_1, N_2 \in \mathbb{N}$ such that

$$n > N_1 \implies |x_n - a| < \frac{\epsilon}{2M}$$

and

$$n > N_2 \implies |y_n - b| < \frac{\epsilon}{2(|a| + 1)}.$$

For $n > \max\{N_1, N_2\}$,

$$\begin{aligned} |x_n y_n - ab| &\leq |x_n y_n - a y_n| + |a y_n - ab| \\ &= |y_n| |x_n - a| + |a| |y_n - b| \\ &< M \frac{\epsilon}{2M} + |a| \frac{\epsilon}{2(|a| + 1)} \\ &< \epsilon. \end{aligned}$$

Therefore $x_n y_n \rightarrow ab$.

- (4) Suppose that $b \neq 0$. Since $y_n \rightarrow b$, there exists $N_0 \in \mathbb{N}$ such that

$$n > N_0 \implies |y_n - b| < \frac{|b|}{2}.$$

The reverse triangle inequality then gives

$$|y_n| \geq |b| - |y_n - b| > \frac{|b|}{2} > 0$$

for every $n > N_0$. In particular, $y_n \neq 0$ eventually. Choose $N_1 \in \mathbb{N}$ such that

$$n > N_1 \implies |y_n - b| < \frac{\epsilon|b|^2}{2}.$$

For $n > \max\{N_0, N_1\}$,

$$\left| \frac{1}{y_n} - \frac{1}{b} \right| = \frac{|y_n - b|}{|y_n||b|} < \frac{2}{|b|^2} |y_n - b| < \epsilon.$$

Hence $1/y_n \rightarrow 1/b$; the finitely many terms with $n \leq N_0$ may be defined arbitrarily because they do not affect the limit. Applying the product law to (x_n) and $(1/y_n)$ yields

$$\frac{x_n}{y_n} \rightarrow \frac{a}{b}.$$

□

2.4.3. Limits and order.

Theorem 2.25. *Suppose that $x_n \rightarrow x$. Then*

- (1) *if $x_n \leq M$ for every $n \in \mathbb{N}$, then $x \leq M$;*
- (2) *if $x_n \geq m$ for every $n \in \mathbb{N}$, then $x \geq m$.*

Proof. (1) Suppose for contradiction that $x > M$, and set

$$\epsilon = \frac{x - M}{2} > 0.$$

Since $x_n \rightarrow x$, there exists $N \in \mathbb{N}$ such that, for every $n > N$,

$$|x_n - x| < \epsilon.$$

It follows that

$$x_n > x - \epsilon = x - \frac{x - M}{2} = \frac{x + M}{2} > M,$$

contradicting $x_n \leq M$. Therefore $x \leq M$.

- (2) Suppose for contradiction that $x < m$, and set

$$\epsilon = \frac{m - x}{2} > 0.$$

Since $x_n \rightarrow x$, there exists $N \in \mathbb{N}$ such that, for every $n > N$,

$$|x_n - x| < \epsilon.$$

It follows that

$$x_n < x + \epsilon = x + \frac{m - x}{2} = \frac{x + m}{2} < m,$$

contradicting $x_n \geq m$. Therefore $x \geq m$.

□

Theorem 2.26 (Squeeze theorem). *Suppose that*

$$x_n \leq y_n \leq z_n$$

for every $n \in \mathbb{N}$. If $x_n \rightarrow a$ and $z_n \rightarrow a$, then $y_n \rightarrow a$.

Proof. Let $\epsilon > 0$. Since $x_n \rightarrow a$, there exists $N_x \in \mathbb{N}$ such that

$$n > N_x \implies a - \epsilon < x_n < a + \epsilon.$$

Similarly, since $z_n \rightarrow a$, there exists $N_z \in \mathbb{N}$ such that

$$n > N_z \implies a - \epsilon < z_n < a + \epsilon.$$

Therefore, for every $n > \max \{N_x, N_z\}$,

$$a - \epsilon < x_n \leq y_n \leq z_n < a + \epsilon.$$

Hence $|y_n - a| < \epsilon$, so $y_n \rightarrow a$. $\square$

The inequalities in the squeeze theorem need only hold for all sufficiently large n , since changing finitely many terms does not affect convergence.

Example 2.27. Consider the oscillating sequence

$$x_n = \frac{(-1)^n}{n}, \quad n \geq 1.$$

Its sign alternates, but

$$-\frac{1}{n} \leq \frac{(-1)^n}{n} \leq \frac{1}{n}.$$

Both bounding sequences converge to 0. Indeed, given $\epsilon > 0$, choosing $N > 1/\epsilon$ gives $1/n < \epsilon$ whenever $n > N$. The squeeze theorem therefore gives

$$\lim_{n \rightarrow \infty} \frac{(-1)^n}{n} = 0.$$

Notice that the squeeze theorem lets us prove convergence without trying to handle even and odd indices separately.

Example 2.28. Consider

$$x_n = \sqrt{n^2 + n} - n, \quad n \geq 1.$$

Rationalising the expression gives

$$x_n = \frac{n}{\sqrt{n^2 + n} + n}.$$

Since

$$n < \sqrt{n^2 + n} < n + 1,$$

we obtain

$$\frac{n}{2n + 1} < x_n < \frac{1}{2}.$$

By the algebraic limit laws,

$$\lim_{n \rightarrow \infty} \frac{n}{2n + 1} = \lim_{n \rightarrow \infty} \frac{1}{2 + 1/n} = \frac{1}{2}.$$

The upper bound is the constant sequence $1/2$, so both bounds converge to $1/2$. Hence the squeeze theorem yields

$$\lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n) = \frac{1}{2}.$$

2.4.4. Subsequences.

Definition 2.29. Let (x_n) be a sequence, and let (n_k) be a strictly increasing sequence of natural numbers:

$$n_0 < n_1 < n_2 < \dots$$

The sequence (x_{n_k}) , obtained by selecting the terms of (x_n) whose indices are $n_0, n_1, n_2, \dots$, is called a *subsequence* of (x_n) .

Theorem 2.30. If $x_n \rightarrow x$, then every subsequence of (x_n) converges to x .

Proof. Let (x_{n_k}) be a subsequence of (x_n) , and let $\epsilon > 0$. Since $x_n \rightarrow x$, there exists $N \in \mathbb{N}$ such that

$$n > N \implies |x_n - x| < \epsilon.$$

Because (n_k) is strictly increasing, $n_k \geq k$ for every $k \in \mathbb{N}$. Thus, whenever $k > N$, we also have $n_k > N$, and hence

$$|x_{n_k} - x| < \epsilon.$$

Therefore $x_{n_k} \rightarrow x$. Since the subsequence was arbitrary, every subsequence of (x_n) converges to the same limit x . $\square$

2.5. 15.09.2026.

2.5.1. Upper and lower bounds.

Definition 2.31. Let $\emptyset \neq E \subseteq \mathbb{R}$. A number $s \in \mathbb{R}$ is an *upper bound* of E if

$$x \leq s \quad \text{for every } x \in E.$$

If such a number exists, E is *bounded above*.

Definition 2.32. Let $\emptyset \neq E \subseteq \mathbb{R}$. A number $p \in \mathbb{R}$ is a *lower bound* of E if

$$p \leq x \quad \text{for every } x \in E.$$

If such a number exists, E is *bounded below*.

Definition 2.33. Let $\emptyset \neq E \subseteq \mathbb{R}$ be bounded above. An upper bound s of E is the *least upper bound*, or *supremum*, of E if $s \leq u$ for every upper bound u of E . We write

$$s = \sup E.$$

Definition 2.34. Let $\emptyset \neq E \subseteq \mathbb{R}$ be bounded below. A lower bound p of E is the *greatest lower bound*, or *infimum*, of E if $p \geq \ell$ for every lower bound ℓ of E . We write

$$p = \inf E.$$

If E is not bounded above, we write $\sup E = +\infty$. Similarly, if E is not bounded below, we write $\inf E = -\infty$.

2.5.2. The completeness axiom.

Completeness axiom. Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

Corollary 2.35. Every nonempty subset $E \subseteq \mathbb{R}$ that is bounded below has an infimum in $\mathbb{R}$.

Proof. Let c be a lower bound of E , and define

$$F = \{-x : x \in E\}.$$

Since $c \leq x$ for every $x \in E$, we have $-x \leq -c$. Thus F is bounded above. It is also nonempty, so the completeness axiom gives $\sup F \in \mathbb{R}$. For every $x \in E$,

$$-x \leq \sup F \leq -c, \quad \text{hence} \quad x \geq -\sup F \geq c.$$

Therefore $-\sup F$ is a lower bound of E . Since the argument applies to every lower bound c of E , it is the greatest lower bound. Consequently,

$$\inf E = -\sup F \in \mathbb{R}.$$

$\square$

2.5.3. Maximum and properties of the supremum.

Definition 2.36. A number s is the *greatest element*, or *maximum*, of a set E if $s \in E$ and $x \leq s$ for every $x \in E$. We write $s = \max E$.

Theorem 2.37. Let $\emptyset \neq E \subseteq \mathbb{R}$ be bounded above. If $s = \max E$, then $s = \sup E$.

Proof. Since $x \leq s$ for every $x \in E$, the number s is an upper bound of E . Hence $\sup E \leq s$. On the other hand, $s \in E$, so $s \leq \sup E$. Thus $s = \sup E$. $\square$

Theorem 2.38. The supremum of a nonempty subset of $\mathbb{R}$ that is bounded above is unique.

Proof. Suppose that s and s' are both least upper bounds of E . Since s' is an upper bound and s is a least upper bound, $s \leq s'$. Similarly, since s is an upper bound and s' is a least upper bound, $s' \leq s$. Therefore $s = s'$. $\square$

Theorem 2.39. Let $\emptyset \neq E \subseteq \mathbb{R}$ be bounded above. For every $\epsilon > 0$, there exists $x \in E$ such that

$$\sup E - \epsilon < x \leq \sup E.$$

Proof. Since $\sup E$ is an upper bound, $x \leq \sup E$ for every $x \in E$. Suppose, for a contradiction, that for some $\epsilon > 0$,

$$x \leq \sup E - \epsilon \quad \text{for every } x \in E.$$

Then $\sup E - \epsilon$ is an upper bound of E , but $\sup E - \epsilon < \sup E$. This contradicts the fact that $\sup E$ is the least upper bound. Hence the claimed element x exists. $\square$

2.5.4. Monotone sequences.

Definition 2.40. A real sequence (x_n) is

- *increasing* (nondecreasing) if $x_n \leq x_{n+1}$ for every $n \in \mathbb{N}$;
- *decreasing* (nonincreasing) if $x_{n+1} \leq x_n$ for every $n \in \mathbb{N}$;
- *strictly increasing* if $x_n < x_{n+1}$ for every $n \in \mathbb{N}$;
- *strictly decreasing* if $x_{n+1} < x_n$ for every $n \in \mathbb{N}$.

A sequence is *monotone* if it is increasing or decreasing, and *strictly monotone* if it is strictly increasing or strictly decreasing.

2.6. 17.09.2026.

2.6.1. Monotone convergence.

Theorem 2.41 (Monotone convergence theorem). Let (x_n) be an increasing sequence of real numbers. If there is $M \in \mathbb{R}$ such that $x_n \leq M$ for every $n \in \mathbb{N}$, then (x_n) converges, and

$$\lim_{n \rightarrow \infty} x_n = \sup \{x_n : n \in \mathbb{N}\} \leq M.$$

Proof. The set $E = \{x_n : n \in \mathbb{N}\}$ is nonempty and bounded above by M . By completeness, $s = \sup E$ exists in $\mathbb{R}$, and $s \leq M$. Let $\epsilon > 0$. By the approximation property of the supremum, there is an index $N \in \mathbb{N}$ such that

$$s - \epsilon < x_N \leq s.$$

Since (x_n) is increasing, for every $n > N$,

$$s - \epsilon < x_N \leq x_n \leq s.$$

Hence $|x_n - s| = s - x_n < \epsilon$, proving $x_n \rightarrow s$. $\square$

Corollary 2.42 (Decreasing case). *Let (x_n) be a decreasing sequence of real numbers. If there is $m \in \mathbb{R}$ such that $m \leq x_n$ for every $n \in \mathbb{N}$, then (x_n) converges, and*

$$\lim_{n \rightarrow \infty} x_n = \inf \{x_n : n \in \mathbb{N}\} \geq m.$$

Consequently, every bounded monotone real sequence converges.

Proof. The sequence $(-x_n)$ is increasing and bounded above by $-m$. By theorem 2.41,

$$-x_n \longrightarrow \sup \{-x_n : n \in \mathbb{N}\}.$$

Multiplying by -1 and using the relation between infima and suprema gives

$$x_n \longrightarrow -\sup \{-x_n : n \in \mathbb{N}\} = \inf \{x_n : n \in \mathbb{N}\} \geq m.$$

A bounded monotone sequence satisfies the hypotheses of either the increasing or the decreasing case, so it converges. $\square$

2.6.2. *Examples of decreasing sequences.* In the following two examples the indices start at $n = 1$.

Example 2.43. Let

$$x_1 = 3, \quad x_n = \frac{n}{n-1} \quad (n \geq 2).$$

Show that (x_n) is strictly decreasing and find its limit.

Solution. First, $x_2 = 2 < 3 = x_1$. For $n \geq 2$, all terms are positive and

$$\frac{x_{n+1}}{x_n} = \frac{(n+1)(n-1)}{n^2} = 1 - \frac{1}{n^2} < 1.$$

Thus $x_{n+1} < x_n$ for every $n \geq 1$. Also $x_1 > 1$, and

$$x_n = 1 + \frac{1}{n-1} > 1 \quad (n \geq 2).$$

Hence the sequence is bounded below and converges by corollary 2.42. We identify its limit in two ways.

Method 1: the squeeze theorem. For $n \geq 2$, we have $n-1 \geq n/2$, so

$$1 < x_n = 1 + \frac{1}{n-1} \leq 1 + \frac{2}{n}.$$

Both bounding sequences tend to 1, so $x_n \rightarrow 1$.

Method 2: the infimum. The number 1 is a lower bound of $E = \{x_n : n \geq 1\}$. If $a > 1$, choose an integer $n > 1 + 1/(a-1)$. Then

$$x_n < a \iff \frac{1}{n-1} < a-1 \iff n > 1 + \frac{1}{a-1}.$$

Thus no $a > 1$ is a lower bound of E . Therefore $\inf E = 1$, and the decreasing case of the monotone convergence theorem yields $\lim_{n \rightarrow \infty} x_n = 1$. $\square$

Example 2.44. Let

$$x_n = \frac{(n+1)^2}{n^2+1}, \quad n \geq 1.$$

Show that (x_n) is decreasing and determine its limit.

Solution. Since every term is positive, we can compare successive terms using their ratio:

$$\begin{aligned} \frac{x_{n+1}}{x_n} &= \frac{(n+2)^2(n^2+1)}{((n+1)^2+1)(n+1)^2} \\ &= \frac{n^4+4n^3+5n^2+4n+4}{n^4+4n^3+7n^2+6n+2}. \end{aligned}$$

The denominator exceeds the numerator by

$$2n^2 + 2n - 2 = 2(n^2 + n - 1) > 0 \quad (n \geq 1).$$

Therefore $x_{n+1}/x_n < 1$, so the sequence is strictly decreasing. Moreover,

$$x_n = 1 + \frac{2n}{n^2 + 1} > 1,$$

so it is bounded below and converges to $\inf \{x_n : n \geq 1\}$.

We now show that this infimum is 1. It is already a lower bound. Given $a > 1$, choose an integer $n > \max \{1, 2/(a-1)\}$. Then

$$x_n = 1 + \frac{2n}{n^2 + 1} < 1 + \frac{2}{n} < a.$$

Hence a is not a lower bound, and consequently

$$\lim_{n \rightarrow \infty} x_n = \inf \{x_n : n \geq 1\} = 1.$$

Alternatively, $1 < x_n < 1 + 2/n$ gives the same limit directly by the squeeze theorem. $\square$

Remark 2.45. The restriction $n \geq 1$ matters for monotonicity in the second example. If we also define x_0 by the same formula, then $x_0 = 1 < x_1 = 2$, so the full sequence is not decreasing. Its tail starting at $n = 1$ is strictly decreasing, and its limit is still 1.

2.6.3. The Bolzano–Weierstrass theorem.

Theorem 2.46 (Bolzano–Weierstrass). *Every bounded sequence of real numbers has a convergent subsequence.*

Proof. Let (x_n) be bounded. Choose $m, M \in \mathbb{R}$ such that $m \leq x_n \leq M$ for every $n \in \mathbb{N}$, and define the *tail suprema*

$$a_k = \sup \{x_n : n \geq k\}, \quad k \in \mathbb{N}.$$

Each tail is nonempty and bounded, so completeness ensures that a_k exists and $m \leq a_k \leq M$. Since

$$\{x_n : n \geq k+1\} \subseteq \{x_n : n \geq k\},$$

the number a_k is an upper bound of the smaller tail as well. Therefore $a_{k+1} \leq a_k$. The sequence (a_k) is decreasing and bounded below, so corollary 2.42 gives a limit

$$a = \lim_{k \rightarrow \infty} a_k \in \mathbb{R}.$$

We construct a subsequence $(x_{n_j})_{j \geq 1}$ recursively. By the approximation property of the supremum, choose $n_1 \geq 1$ such that

$$a_1 - 1 < x_{n_1} \leq a_1.$$

Once n_j has been chosen, apply the same property to the tail starting at $n_j + 1$. There is an index $n_{j+1} \geq n_j + 1$ such that

$$a_{n_j+1} - \frac{1}{j+1} < x_{n_{j+1}} \leq a_{n_j+1}.$$

Thus the indices are strictly increasing, as required for a subsequence.

To verify convergence, observe that $a_{n_j+1} \leq a_{n_j+1}$ and $x_{n_{j+1}} \leq a_{n_j+1}$. Consequently, for every $j \geq 1$,

$$a_{n_j} - \frac{1}{j} < x_{n_j} \leq a_{n_j}.$$

For $j = 1$, the lower inequality follows from $a_{n_1} \leq a_1$; for later indices it follows from the recursive choice above. Since (a_{n_j}) is a subsequence of the convergent sequence (a_k) , it tends to a . Also $1/j \rightarrow 0$, so both bounding sequences tend to a . The squeeze theorem now yields $x_{n_j} \rightarrow a$, proving the claim. $\square$

2.7. 22.09.2026.

2.7.1. Bernoulli's inequality.

Theorem 2.47 (Bernoulli's inequality). *For every $x \in \mathbb{R}$ with $x > -1$ and every $n \in \mathbb{N}$,*

$$(1+x)^n \geq 1+nx.$$

Proof. Fix $x > -1$. The cases $n = 0$ and $n = 1$ are equalities. Suppose that $(1+x)^n \geq 1+nx$ for some $n \geq 1$. Since $1+x > 0$, multiplication preserves the inequality, and

$$\begin{aligned} (1+x)^{n+1} &= (1+x)(1+x)^n \\ &\geq (1+x)(1+nx) \\ &= 1+(n+1)x+nx^2 \\ &\geq 1+(n+1)x. \end{aligned}$$

The result follows by induction. If $x \neq 0$, then $nx^2 > 0$ in the induction step, so the inequality is strict for $n \geq 2$. $\square$

2.7.2. A sequence defining the number e . Consider the sequence

$$x_n = \left(1 + \frac{1}{n}\right)^n, \quad n \geq 1.$$

Lemma 2.48. *The sequence (x_n) is strictly increasing.*

Proof. All terms are positive. For $n \geq 1$,

$$\begin{aligned} \frac{x_{n+1}}{x_n} &= \frac{\left(1 + \frac{1}{n+1}\right)^{n+1}}{\left(1 + \frac{1}{n}\right)^n} \\ &= \left(\frac{1 + \frac{1}{n+1}}{1 + \frac{1}{n}}\right)^{n+1} \left(1 + \frac{1}{n}\right) \\ &= \left(\frac{n(n+2)}{(n+1)^2}\right)^{n+1} \frac{n+1}{n} \\ &= \left(1 - \frac{1}{(n+1)^2}\right)^{n+1} \frac{n+1}{n} \\ &> \left(1 - \frac{n+1}{(n+1)^2}\right) \frac{n+1}{n} = 1. \end{aligned}$$

The strict inequality follows from theorem 2.47, since $-1 < -1/(n+1)^2 < 0$ and $n+1 \geq 2$. Hence $x_{n+1} > x_n$. $\square$

Lemma 2.49. *The sequence (x_n) is bounded above.*

Proof. We use the auxiliary sequence

$$y_n = \left(1 - \frac{1}{n}\right)^n, \quad n \geq 2.$$

This sequence is positive and strictly increasing. Indeed, by theorem 2.47, for $n \geq 2$,

$$\begin{aligned} \frac{y_{n+1}}{y_n} &= \left(\frac{n^2}{n^2 - 1} \right)^n \frac{n}{n+1} \\ &= \left(1 + \frac{1}{n^2 - 1} \right)^n \frac{n}{n+1} \\ &\geq \left(1 + \frac{n}{n^2 - 1} \right) \frac{n}{n+1} \\ &= 1 + \frac{1}{(n+1)(n^2 - 1)} > 1. \end{aligned}$$

Consequently, $(1/y_n)$ is decreasing. Also, for $n \geq 2$,

$$\left(1 + \frac{1}{n} \right) \left(1 - \frac{1}{n} \right) = 1 - \frac{1}{n^2} < 1.$$

Dividing by $1 - 1/n > 0$ and raising to the power n gives

$$x_n < \left(1 - \frac{1}{n} \right)^{-n} = \frac{1}{y_n} \leq \frac{1}{y_2} = \left(1 - \frac{1}{2} \right)^{-2} = 4.$$

Finally, $x_1 = 2 < 4$, so the bound holds for every $n \geq 1$. □

By theorem 2.41, the increasing, bounded sequence (x_n) converges to a real number.

Definition 2.50. The limit

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n$$

is called *Euler's number* (*Neperin luku* in Finnish).